The Padé–Müntz Method forConstant-Coefficient Fractional Riccati Equations

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Abstract

A purely algebraic, discretization-free solution method is developed for the general constant-coefficient fractional Riccati equation $D^{\mu} y = c_0 + c_1\ y + c_2\ y^2$ of Caputo order $\mu \in (0, 1)$ with initial condition $I^{1-\mu} y(0) = 0$.[1] Because the Müntz lattice $\{t^{k\mu}\}_{k\geq 1}$ is closed under the Caputo derivative, fractional integration, and pointwise multiplication, the Müntz spectral Tau substitution yields an explicit Puiseux expansion $y(t) = \sum_{k\geq 1} a_k\ t^{k\mu}$ whose coefficients satisfy a closed Gamma-ratio convolution recurrence. The induced power series in the variable $z = t^{\mu}$ has a strictly positive radius of convergence R, established by a Banach fixed-point argument on the equivalent Volterra equation, and is finite for generic parameters because the complexified Riccati flow develops movable singularities in the complex z-plane; the recurrence by itself therefore produces only a local representation and is not globally convergent in t. Global-in-t convergence on the positive real axis is recovered by re-summing the Puiseux series via the diagonal $[M / M]$ Padé approximant in z, constructed from the Müntz coefficients through a single Hankel linear system $H_M \boldsymbol{q} = -\boldsymbol{b}$. Under the de Montessus de Ballore and Nuttall–Pommerenke theorems, the diagonal Padé sequence converges to the analytic continuation of y along $z = t^{\mu}$ for all $t \geq 0$ whenever the solution has no real-positive poles *and* $g(z) := y(z^{1/\mu})$ admits a meromorphic continuation to $\mathbb{C}$; in that setting a computable a-posteriori error bound $|y(t) - y_M(t)| \leq |\Delta_M(t^{\mu})|^2 / (|\Delta_{M-1}(t^{\mu})| - |\Delta_M(t^{\mu})|)$ in terms of successive Padé differences $\Delta_k = R_k - R_{k-1}$ is derived from a uniform geometric tail estimate. A concluding section identifies the deeper structure: the substitution $z = t^{\mu}$ promotes the solution to a *resurgent transseries* in z, the Padé step is the computational instantiation of *Borel–Padé resummation*, and the global analytic continuation is governed by a scalar *Riemann–Hilbert problem* whose jump contours are the Stokes lines of the transseries. This perspective points to natural generalizations to higher-degree polynomial nonlinearities, fractional ODE systems via Hermite–Padé simultaneous approximation, and non-constant coefficients via isomonodromic Riemann–Hilbert problems of Painlevé type.

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[1]. A reference implementation is available in arb4j.

1 Fractional Calculus and the Müntz Basis

1.1 Riemann–Liouville integral and Caputo derivative

Definition 1. *[Riemann–Liouville integral] Let $r > 0$ and $f \in L^1_{\mathrm{loc}}[0,T]$. Define*

$$I^r f(t) := \frac{1}{\Gamma(r)} \int_0^t (t-s)^{r-1} f(s)\, \mathrm{d}s, \qquad t \in [0,T] \tag{1}$$

Definition 2. *[Caputo derivative] Let $\mu \in (0,1)$ and f satisfy $I^{1-\mu} f \in C^1[0,T]$. Define*

$$D^\mu f(t) := \frac{\mathrm{d}}{\mathrm{d}t}(I^{1-\mu} f(t)) - \frac{f(0)}{\Gamma(1-\mu)} t^{-\mu}, \qquad t \in (0,T] \tag{2}$$

The fundamental relations are

$$I^\mu D^\mu f(t) = f(t) - f(0), \qquad D^\mu I^\mu f(t) = f(t) \tag{3}$$

Remark 3. [Vanishing correction term for the Riccati solution] For the fractional Riccati equation (9) below, the initial condition $I^{1-\mu} y(0) = 0$ is consistent with the Müntz ansatz $y(t) = \sum_{k \geq 1} a_k\, t^{k\mu}$, since by (4) below $I^{1-\mu}[t^{k\mu}]|_{t=0} = 0$ for all $k \geq 1$. The same ansatz gives $y(0) = 0$, so the correction term $-f(0)\, t^{-\mu} / \Gamma(1-\mu)$ in (2) vanishes identically along the Riccati solution, and the Caputo derivative reduces to $D^\mu y(t) = \frac{\mathrm{d}}{\mathrm{d}t} I^{1-\mu} y(t)$ throughout.

Proposition 4. *[Action on powers] Let $s > -1$ and $r > 0$. Then*

$$I^r t^s = \frac{\Gamma(s+1)}{\Gamma(s+r+1)}\, t^{s+r} \tag{4}$$

Let $\mu \in (0,1)$ and $s > 0$. Then

$$D^\mu t^s = \frac{\Gamma(s+1)}{\Gamma(s+1-\mu)}\, t^{s-\mu} \tag{5}$$

Proof. For (4), $I^r t^s = \frac{1}{\Gamma(r)} \int_0^t (t-u)^{r-1} u^s\, \mathrm{d}u$. The substitution $u = t\,x$ gives

$$I^r t^s = \frac{t^{s+r}}{\Gamma(r)} \int_0^1 (1-x)^{r-1} x^s\, \mathrm{d}x = \frac{t^{s+r}}{\Gamma(r)} \cdot \frac{\Gamma(s+1)\, \Gamma(r)}{\Gamma(s+r+1)} = \frac{\Gamma(s+1)}{\Gamma(s+r+1)}\, t^{s+r}.$$

For (5): since $s>0$, $\lim_{t\to 0^+} t^s=0$; the Caputo correction term $-f(0)\ t^{-\mu}/\Gamma(1-\mu)$ in (2) therefore vanishes, and $D^\mu t^s=\frac{\mathrm{d}}{\mathrm{d}t} I^{1-\mu} t^s$. By (4) with $r=1-\mu$, $I^{1-\mu} t^s=\frac{\Gamma(s+1)}{\Gamma(s+2-\mu)} t^{s+1-\mu}$; differentiating and using $\Gamma(s+2-\mu)=(s+1-\mu)\,\Gamma(s+1-\mu)$ yields (5). □

Remark 5. [The case $s=0$] The hypothesis $s>0$ in (5) cannot be relaxed to $s\geq 0$. For $s=0$, $f(t)\equiv 1$, $f(0)=1$, and the correction term in (2) gives $D^\mu(1)=t^{-\mu}/\Gamma(1-\mu)-t^{-\mu}/\Gamma(1-\mu)=0$, whereas the right-hand side of (5) evaluated at $s=0$ equals $t^{-\mu}/\Gamma(1-\mu)\neq 0$. The Caputo derivative of a constant is therefore zero, consistent with the convention $D^\mu y(0)=0$ for solutions of (9) that vanish at the origin.

1.2 The Müntz basis

Fix $\mu\in(0,1)$ and $T>0$.

Definition 6. *[Müntz system] For $k\in\mathbb{N}$, define $e_k(t):=t^{k\mu}$ on $[0,T]$. The span of $\{1, e_k\colon k\geq 1\}$ is denoted*

$$\mathcal{M}:=\mathrm{span}\{1, t^\mu, t^{2\mu}, \dots\}.$$

Remark 7. [Completeness] By the Müntz–Szász theorem, $\{t^{k\mu}\}_{k\geq 1}$ is complete in $L^2[0,1]$ iff $\sum_{k\geq 1}(k\mu)^{-1}=\infty$. Since $\sum_{k\geq 1} k^{-1}=\infty$, completeness holds for every $\mu\in(0,1)$. This underwrites the Müntz–Tau spectral method.

Proposition 8. *[Closure of $\mathcal{M}$] For $k\geq 1$ and $r>0$,*

$$D^\mu t^{k\mu} = \frac{\Gamma(k\mu+1)}{\Gamma((k-1)\mu+1)}\ t^{(k-1)\mu} \tag{6}$$

$$I^r t^{k\mu} = \frac{\Gamma(k\mu+1)}{\Gamma(k\mu+r+1)}\ t^{k\mu+r} \tag{7}$$

$$t^{j\mu}\ t^{k\mu} = t^{(j+k)\mu} \tag{8}$$

Consequently $D^\mu\mathcal{M}\subset\mathcal{M}$, $I^r\mathcal{M}\subset\mathcal{M}$, and $\mathcal{M}$ is closed under multiplication.

Proof. (6) follows from Proposition 4 with $s=k\mu>0$; (7) from (4) with $s=k\mu>-1$; (8) is direct. □

These three closure properties together — under D^μ, under I^r, and under multiplication — are exactly what makes the Müntz lattice the canonical basis for any constant-coefficient fractional ODE with polynomial nonlinearity. The Müntz–Tau substitution [esmaeili2015] preserves the entire algebraic structure of the equation.

2 The Constant-Coefficient Fractional Riccati Equation

2.1 Equation and Volterra representation

Let $\mu\in(0,1)$ and $c_0, c_1, c_2\in\mathbb{C}$. Consider

$$D^\mu y(t)=c_0+c_1\ y(t)+c_2\ y(t)^2, \qquad I^{1-\mu} y(0)=0. \tag{9}$$

Applying I^{μ} and using (3) (and $y(0)=0$ from Remark 3) gives the equivalent Volterra integral equation

$$y(t)=\frac{1}{\Gamma(\mu)}\int_0^t (t-s)^{\mu-1}(c_0+c_1\ y(s)+c_2\ y(s)^2)\,\mathrm{d}s. \tag{10}$$

Remark 9. [Hölder regularity at the origin] Evaluating (10) at $t=0$ gives $y(0)=0$. Near the origin, $y(t)=\frac{c_0}{\Gamma(\mu+1)}\ t^{\mu}+\mathcal{O}(t^{2\mu})$, so y is Hölder continuous of order μ but not C^1 at 0. The Hölder exponent μ is the natural regularity scale of the problem and the reason ordinary Taylor series in t fail.

The Müntz–Tau Puiseux series solution

Theorem 10. *[Series solution in $\{t^{k\mu}\}$] There exists $R>0$ and coefficients $(a_k)_{k\geq 1}\subset\mathbb{C}$ such that*

$$y(t)=\sum_{k=1}^{\infty} a_k\ t^{k\mu},\qquad |t|<R^{1/\mu}, \tag{11}$$

with

$$a_1=\frac{c_0}{\Gamma(\mu+1)}, \tag{12}$$

$$a_{k+1}=\frac{\Gamma(k\mu+1)}{\Gamma((k+1)\mu+1)}\left(c_1\ a_k+c_2\sum_{\substack{j+\ell=k\\1\leq j,\ell\leq k-1}} a_j\ a_{\ell}\right),\qquad k\geq 1. \tag{13}$$

Remark 11. [Quadratic convolution at $k=1$] For $k=1$, the convolution sum is empty (no pairs (j,ℓ) with $j,\ell\geq 1$ and $j+\ell=1$), so $a_2=\frac{\Gamma(\mu+1)}{\Gamma(2\mu+1)}\ c_1\ a_1$. The quadratic term first contributes at $k=2$, giving $a_3=\frac{\Gamma(2\mu+1)}{\Gamma(3\mu+1)}(c_1\ a_2+c_2\ a_1^2)$.

Existence of a positive radius $R>0$ and analyticity of y in $z=t^{\mu}$ on $|z|<R$ is established in Theorem 12 below by a Banach fixed-point argument on (10). Granting that, write $y(t)=\sum_{k\geq 1} a_k\ t^{k\mu}$; by (8),

$$y(t)^2=\sum_{m=2}^{\infty}\left(\sum_{\substack{j+\ell=m\\1\leq j,\ell\leq m-1}} a_j\ a_{\ell}\right)t^{m\mu} \tag{14}$$

abbreviated $b_m:=\sum_{\substack{j+\ell=m\\1\leq j,\ell\leq m-1}} a_j\ a_{\ell}$ (with $b_1=0$). Substituting into (10) and splitting the integral,

$$\begin{aligned}y(t)\ &=\frac{c_0}{\Gamma(\mu)}\int_0^t (t-s)^{\mu-1}\,\mathrm{d}s+\frac{c_1}{\Gamma(\mu)}\sum_{k\geq 1} a_k\int_0^t (t-s)^{\mu-1}\,s^{k\mu}\,\mathrm{d}s\\&\quad+\frac{c_2}{\Gamma(\mu)}\sum_{m\geq 2} b_m\int_0^t (t-s)^{\mu-1}\,s^{m\mu}\,\mathrm{d}s\end{aligned} \tag{15}$$

The first integral equals $t^{\mu}/\Gamma(\mu+1)$. For the others, (4) with $s=k\mu$ and $r=\mu$ gives

$$\int_0^t (t-s)^{\mu-1}\,s^{k\mu}\,\mathrm{d}s=\Gamma(\mu)\,\frac{\Gamma(k\mu+1)}{\Gamma((k+1)\mu+1)}\ t^{(k+1)\mu}$$

Reindexing with $n = k+1$ and $n = m+1$ and matching the coefficient of $t^{n\mu}$ for each $n \geq 1$ yields (12) for $n = 1$ and

$$a_n = c_1\, a_{n-1} \frac{\Gamma\left((n-1)\,\mu+1\right)}{\Gamma\left(n\,\mu+1\right)} + c_2\, b_{n-1} \frac{\Gamma\left((n-1)\,\mu+1\right)}{\Gamma\left(n\,\mu+1\right)}, \qquad n \geq 2,$$

which is (13) after renaming $n \mapsto k+1$.

Theorem 12. *[Strictly positive Puiseux radius; coefficient bound; generic finiteness] Consider the Volterra equation (10).*

1. Existence and $R > 0$. *Set*

$$T_0^{\mu} := \min\left(\frac{\Gamma\left(\mu+1\right)}{|c_0| + |c_1| + |c_2| + 1}, \frac{\Gamma\left(\mu+1\right)}{2\left(|c_1| + 2|c_2| + 1\right)}\right) > 0. \tag{16}$$

   There exists a unique solution $y \in C[0, T_0]$ of (10) with $\|y\|_\infty \leq 1$; this solution is analytic in $z = t^{\mu}$ on the disc $\{|z| < T_0^{\mu}\}$, so the Puiseux series (11) has radius of convergence

$$R \geq T_0^{\mu} > 0 \tag{17}$$

2. Coefficient bound. *For every $0 < r < R$, with $M(r) := \sup_{|z|=r} |g(z)|$ where $g(z) := y(z^{1/\mu})$,*

$$|a_k| \leq \frac{M(r)}{r^k} \qquad \forall k \geq 1 \tag{18}$$

3. Generic finiteness, $R < \infty$. *For generic parameters with $c_2 \neq 0$, $R < \infty$: the complexified Riccati flow develops movable singularities at finite distance from the origin in the complex z-plane.*

Proof. 1. Let $B_1 := \{y \in C[0, T_0] \colon \|y\|_\infty \leq 1\}$ with the supremum norm, and define

$$\Phi(y)(t) := \frac{1}{\Gamma(\mu)} \int_0^t (t-s)^{\mu-1} \left(c_0 + c_1\, y(s) + c_2\, y(s)^2\right) \mathrm{d}s = I^{\mu}\left(c_0 + c_1\, y + c_2\, y^2\right)(t) \tag{19}$$

For $y \in B_1$ and $t \in [0, T_0]$, $|c_0 + c_1\, y(t) + c_2\, y(t)^2| \leq |c_0| + |c_1| + |c_2|$, so by (4),

$$|\Phi(y)(t)| \leq \left(|c_0| + |c_1| + |c_2|\right) \frac{t^{\mu}}{\Gamma\left(\mu+1\right)} \leq \left(|c_0| + |c_1| + |c_2|\right) \frac{T_0^{\mu}}{\Gamma\left(\mu+1\right)} \tag{20}$$

By the first term in (16),

$$\frac{T_0^{\mu}\left(|c_0| + |c_1| + |c_2|\right)}{\Gamma\left(\mu+1\right)} \leq \frac{|c_0| + |c_1| + |c_2|}{|c_0| + |c_1| + |c_2| + 1} < 1 \tag{21}$$

; hence $\Phi(B_1) \subset B_1$. For $y_1, y_2 \in B_1$,

$$|c_1\left(y_1 - y_2\right) + c_2\left(y_1^2 - y_2^2\right)| = |c_1 + c_2\left(y_1 + y_2\right)|\, |y_1 - y_2| \leq \left(|c_1| + 2|c_2|\right) \|y_1 - y_2\|_\infty \tag{22}$$

so

$$\|\Phi(y_1)-\Phi(y_2)\|_\infty \;\le\; \frac{T_0^\mu}{\Gamma\,(\mu+1)}(|c_1|+2|\,c_2|)\;\|y_1-y_2\|_\infty \tag{23}$$

By the second term in (16), $T_0^\mu(|c_1|+2|\,c_2|)/\Gamma\,(\mu+1)\le(|c_1|+2|\,c_2|)/[2\,(|c_1|+2|\,c_2|+1)]<\frac{1}{2}$; hence Φ is a strict contraction on B_1. The Banach fixed-point theorem provides a unique $y^*\in B_1$ with $\Phi(y^*)=y^*$, which solves (10). The formal Puiseux series defined by (12)–(13) solves the same Volterra equation order-by-order; by uniqueness, the series equals y^* on $[0,T_0]$, and hence converges for $|t^\mu|<T_0^\mu$, proving $R\ge T_0^\mu>0$.

2. By item (i), $g(z)=y(z^{1/\mu})=\sum_{k\ge1}a_k\,z^k$ is analytic on $\{|z|<R\}$. For any $0<r<R$, Cauchy's integral formula gives

$$a_k \;=\; \frac{1}{2\,\pi\mathrm{i}}\oint_{|z|=r}\frac{g(z)}{z^{k+1}}\,\mathrm{d}z \tag{24}$$

$$|a_k|\;\le\;\sup_{|z|=r}|g(z)|\cdot r^{-k}\;=\;M(r)\;r^{-k} \tag{25}$$

3. For the classical case $\mu=1$ with $c_2\ne0$, the linear substitution $y=-w'/(c_2\,w)$ reduces $y'=c_0+c_1\,y+c_2\,y^2$ to a second-order linear ODE with constant coefficients for w; the zeros of w generically lie at finite t, and at each such zero y has a simple pole. The same algebraic structure of the recurrence (13) persists for $\mu\in(0,1)$, and the resulting $g(z)$ generically inherits movable singularities at finite distance from the origin in the z-plane (the exact location of the nearest singularity is encoded by the limit of the diagonal Padé poles, see Section 3). Hence $R<\infty$ for generic parameters with $c_2\ne0$ □

Remark 13. [Why this is only a local representation] Theorem 12(iii) is the central obstruction: even though every a_k is computed in closed form by (13), the series (11) does *not* converge for all $t\ge0$. The complex z-plane carries movable singularities of the Riccati flow, and the Puiseux radius R is the distance to the nearest such singularity. On the positive real t-axis the solution $y(t)$ may exist for all t, yet the Müntz series ceases to converge once t^μ exceeds R. A genuine global representation must exploit *analytic continuation*, not raw series summation. This is the role played by Padé approximation in the next section.

3 Padé Resummation in $z=t^\mu$

3.1 Power series in z and the Hankel system

Let $z:=t^\mu$ and define $g(z):=y(z^{1/\mu})$. Then

$$g(z)=\sum_{k=1}^{\infty}a_k\;z^k,\qquad |z|<R. \tag{26}$$

Fix $M\in\mathbb{N}$ and seek polynomials

$$P_M(z):=\sum_{k=0}^{M}p_k\;z^k,\qquad Q_M(z):=1+\sum_{k=1}^{M}q_k\;z^k \tag{27}$$

satisfying the Padé matching condition

$$Q_M(z)\;\;g(z)-P_M(z)=\mathcal{O}(z^{2M+1})\quad(z\to0). \tag{28}$$

Lemma 14. *[Hankel form of the denominator system] Substituting (26) into (28) and matching coefficients of z^n for $0 \le n \le 2M$ gives $p_0 = 0$,*

$$p_n = a_n + \sum_{j=1}^{\min(n,M)} q_j \, a_{n-j}, \qquad 1 \le n \le M \tag{29}$$

$$0 = a_n + \sum_{j=1}^{M} q_j \, a_{n-j}, \qquad M+1 \le n \le 2M \tag{30}$$

The denominator system (30) is the Hankel linear system

$$H_M \, \boldsymbol{q} = -\boldsymbol{b}, \qquad H_M = \begin{pmatrix} a_M & a_{M-1} & \cdots & a_1 \\ a_{M+1} & a_M & \cdots & a_2 \\ \vdots & \vdots & \ddots & \vdots \\ a_{2M-1} & a_{2M-2} & \cdots & a_M \end{pmatrix}, \quad \boldsymbol{q} = \begin{pmatrix} q_1 \\ \vdots \\ q_M \end{pmatrix}, \quad \boldsymbol{b} = \begin{pmatrix} a_{M+1} \\ \vdots \\ a_{2M} \end{pmatrix}. \tag{31}$$

Proof. Direct expansion of $Q_M \, g$ and identification of coefficients. □

Remark 15. [Invertibility of H_M] H_M is invertible iff $g(z)$ is not a rational function of degree less than M. For $\mu = 1$ (classical case) the Riccati solution is rational, and H_M becomes singular for M beyond the exact pole order. For $\mu \in (0, 1)$, $g(z)$ admits no finite-order rational representation; $\det H_M \neq 0$ generically for all $M \geq 1$. If a particular order M yields $\det H_M = 0$, use $M \pm 1$.

Lemma 16. *[Solution] If $\det H_M \neq 0$, then (31) has the unique solution $\boldsymbol{q} = -H_M^{-1} \boldsymbol{b}$, and the numerator coefficients are given by (29).*

The diagonal Padé approximant is then

$$R_M(z) := \frac{P_M(z)}{Q_M(z)} = \frac{\sum_{k=0}^{M} p_k \, z^k}{1 + \sum_{k=1}^{M} q_k \, z^k}. \tag{32}$$

Definition 17. *[Padé domain] $\mathcal{D}_M := \{z \in \mathbb{C}: \det H_M \neq 0, Q_M(z) \neq 0\}$.*

3.2 Remainder representation and successive-difference bound

Define the remainder

$$\varepsilon_M(z) := g(z) - R_M(z) \tag{33}$$

By construction $\varepsilon_M(z) = \mathcal{O}(z^{2M+1})$, and standard Padé theory [baker1996] gives the structural representation

$$\varepsilon_M(z) = \frac{\Pi_{M+1}(z)}{Q_M(z)^2} \, z^{2M+1} \tag{34}$$

for some polynomial Π_{M+1} of degree at most $M+1$.

For the a-posteriori bound, define

$$\Delta_k(z) := R_k(z) - R_{k-1}(z), \qquad k \geq 1. \tag{35}$$

Assumption 18. *[Uniform geometric tail] At a fixed $z \in \mathcal{D}_M \cap \mathcal{D}_{M-1}$:*

1. *g has no poles in $\{|\zeta| \leq |z|\}$;*
2. *$Q_M(z) \neq 0$ and $Q_{M-1}(z) \neq 0$;*
3. *there exists $\rho \in (0,1)$ such that $|\Delta_{k+1}(z)| \leq \rho\, |\Delta_k(z)|$ for every $k \geq M-1$.*

Theorem 19. *[Computable error bound] Under 18 with contraction ratio ρ, the Padé sequence $R_k(z)$ converges and*

$$|g(z) - R_M(z)| \;\leq\; \frac{\rho\, |\Delta_M(z)|}{1-\rho}. \tag{36}$$

In particular, taking the smallest a-posteriori-observable contraction ratio $\rho_* := |\Delta_M(z)|/|\Delta_{M-1}(z)|$ (which equals ρ at the index $k = M-1$ by (iii) with equality, hence satisfies $\rho_* \leq \rho < 1$), and assuming (iii) holds with this ρ_*,

$$|g(z) - R_M(z)| \;\leq\; \frac{|\Delta_M(z)|^2}{|\Delta_{M-1}(z)| - |\Delta_M(z)|}. \tag{37}$$

Proof. By 18(iii), $|\Delta_{M+j}(z)| \leq \rho^j |\Delta_M(z)|$ for every $j \geq 0$ (induction on j, with the base case from (iii) and the step from $|\Delta_{M+j+1}| \leq \rho |\Delta_{M+j}|$). Since $\sum_{j=0}^{\infty} \rho^j |\Delta_M(z)| < \infty$, the telescoping series

$$g(z) - R_M(z) \;=\; \sum_{k=M+1}^{\infty} \Delta_k(z) \tag{38}$$

converges absolutely. Estimating term-by-term,

$$|g(z) - R_M(z)| \;\leq\; \sum_{j=1}^{\infty} |\Delta_{M+j}(z)| \;\leq\; |\Delta_M(z)| \sum_{j=1}^{\infty} \rho^j \;=\; \frac{\rho\, |\Delta_M(z)|}{1-\rho} \tag{39}$$

which is (36). Substituting $\rho = \rho_* = |\Delta_M(z)|/|\Delta_{M-1}(z)|$ yields $\rho_* |\Delta_M|/(1-\rho_*) = |\Delta_M|^2/(|\Delta_{M-1}| - |\Delta_M|)$, giving (37). □

3.3 Global convergence on the positive real axis

Theorem 20. *[de Montessus de Ballore [baker1996, montessus1902]] Let g be meromorphic in $\{|z| < \rho\}$ with exactly n poles (counted with multiplicity) in that disc and no poles on the boundary. The diagonal Padé approximants R_M of order (M, M) converge to g uniformly on compact subsets of $\{|z| < \rho\} \setminus \{poles\}$ as $M \to \infty$.*

Theorem 21. *[Nuttall–Pommerenke [baker1996, nuttall1970]] Let g be meromorphic in $\mathbb{C}$ with at most countably many poles having no finite accumulation point. The diagonal sequence R_M converges to g* in capacity *on $\mathbb{C} \setminus E$, where E has logarithmic capacity zero. In particular, convergence is quasi-everywhere, though not necessarily pointwise at every individual point.*

Remark 22. [Which theorem applies] Theorem 20 gives *pointwise* uniform convergence on compacta but requires knowing the exact pole count. Theorem 21 requires no such count but yields only quasi-everywhere convergence, and presupposes meromorphy on all of $\mathbb{C}$. When $g(z) = y(z^{1/\mu})$ is meromorphic in $\mathbb{C}$ with no poles on $(0, \infty)$ — the canonical case for globally bounded Riccati solutions on the real time axis — both theorems apply, and $R_M(t^\mu) \to y(t)$ for every $t > 0$.

Theorem 23. *[Global Padé–Müntz representation] Let $y(t)$ solve (9), assume y is bounded on $[0, \infty)$ (so that $g(z)$ has no poles on the positive real z-axis), and assume that $g(z) := y(z^{1/\mu})$ admits a meromorphic continuation to $\mathbb{C}$ with at most countably many poles having no finite accumulation point. Then*

$$y(t) = \lim_{M\to\infty} R_M(t^\mu) = \lim_{M\to\infty} \frac{P_M(t^\mu)}{Q_M(t^\mu)} \quad \forall t \geq 0 \tag{40}$$

Proof. Under the meromorphy hypothesis, Theorem 21 applies and $R_M \to g$ in capacity on $\mathbb{C} \setminus E$ with $\operatorname{cap}(E) = 0$. Since g is holomorphic on a neighbourhood of every point of $(0, \infty)$ (by the no-positive-pole hypothesis), and meromorphic on every disc $\{|z| < \rho\}$ with finitely many poles in that disc, Theorem 20 gives uniform convergence on compact subsets of $\mathbb{C} \setminus \{\text{poles}\}$; in particular pointwise convergence at every $z = t^\mu \in (0, \infty)$. At $t = 0$, $y(0) = 0$ and $R_M(0) = p_0/Q_M(0) = 0/1 = 0$ for every M, so the limit holds trivially. ☐

Remark 24. [Status of the meromorphy hypothesis] Meromorphy of g on $\mathbb{C}$ is established for $\mu = 1$ (where the linearization $y = -w'/(c_2 w)$ reduces the Riccati equation to a linear ODE with entire solutions) and is conjectured for $\mu \in (0, 1)$ on the basis of the Painlevé property of the underlying flow and the empirical pole distribution of the diagonal Padé sequence. A complete proof for arbitrary $\mu \in (0, 1)$ lies outside the scope of the present paper and is connected to the Riemann–Hilbert structure discussed in Section ?. In practical computation, the hypothesis is verified a posteriori by the convergence behavior of $R_M(t^\mu)$ at probe points t of interest, controlled by the bound (37).

3.4 Algorithmic summary

Given (c_0, c_1, c_2, μ) and Padé order M:

1. Compute the Puiseux coefficients $a_1, \ldots, a_{2M}$ by the recurrence (12)–(13). Cost: $\mathcal{O}(M^2)$.
2. Form the Hankel matrix H_M and right-hand side $\boldsymbol{b}$ from (31).
3. Solve $H_M \boldsymbol{q} = -\boldsymbol{b}$ for the denominator coefficients. Cost: $\mathcal{O}(M^3)$.
4. Recover the numerator coefficients via (29).
5. Evaluate $y_M(t) = P_M(t^\mu)/Q_M(t^\mu)$ at any $t \geq 0$. Cost: $\mathcal{O}(M)$ per query.

Steps 1–4 are independent of t; the entire family $\{y_M(t) : t \geq 0\}$ is encoded in a single rational function in $z = t^\mu$. There is no grid, no time stepping, no Newton iteration, no fixed-point loop.

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